GLOBAL STUDY SUPPORT / REFERENCE SERIES

# Python Data Analysis

ILLUSTRATIVE DEMONSTRATION - NOT CLIENT WORK

## Objective

Explore a fictional sales dataset using a transparent, reproducible workflow. The example compares total revenue across three regions and creates a simple bar chart.

## Workflow

1. Load sales.csv with pandas.
2. Verify required columns, missing values and numeric revenue.
3. Group revenue by region.
4. Export a summary and create a labelled chart.
5. Explain the result and the limits of the small dataset.

## Run the example

Install pandas and matplotlib in your Python environment. Place python-analysis.py and sales.csv in the same directory. Run: python python-analysis.py. The script writes regional-summary.csv and regional-revenue.png.

## Interpretation and limitations

North has revenue of 8,200, South 7,300 and West 9,500 in the invented dataset. West contributes 38% of the total 25,000. These six observations are insufficient for forecasting, causal inference or a real business decision. All records are synthetic.

| Region | Revenue (illustrative) | Share |
|---|---|---|
| North | 8,200 | 32.8% |
| South | 7,300 | 29.2% |
| West | 9,500 | 38.0% |